

# Autodesk® Scaleform®

Scaleform 4.3

: Artem Bolgar  
:  
: 2012 9 17

Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk 123D, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo), BIM 360, Built with ObjectARX (design/logo), Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, Design Server, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, Exposure, Extending the Design Team, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, Freewheel, GDX Driver, Glue, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, Map It, Build It, Use It, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder,

Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, Revit LT, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Socialcam, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

---

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

[www.scaleform.com](http://www.scaleform.com)

[info@scaleform.com](mailto:info@scaleform.com)

(301) 446-3200

(301) 446-3199



|     |                  |           |
|-----|------------------|-----------|
|     |                  | 5         |
|     |                  | 3         |
| 2.1 |                  | 3         |
| 2.2 |                  | 3         |
| 2.3 | Mouse            | 5         |
| 2.4 |                  | 7         |
| 2.5 |                  | 9         |
|     |                  | 13        |
|     |                  | 15        |
|     |                  | 19        |
|     |                  | 23        |
|     | <b>TextField</b> | <b>33</b> |
| 7.1 |                  | 33        |
| 7.2 |                  | 43        |
| 7.3 |                  | 50        |
| 7.4 | HTML             | 53        |
| 7.5 | Shadow           | 55        |
| 7.6 |                  | 59        |
| 7.7 | IME              | 63        |
|     |                  | 66        |
|     |                  | 67        |
|     |                  | 70        |
|     |                  | 71        |
|     |                  | 72        |
| 13  |                  | 73        |

|       |    |
|-------|----|
| ..... | 76 |
| ..... | 77 |

```

Autodesk Scaleform SDK
PC
Flash®
Scaleform

,
IME
Scaleform
Scaleform Player
true
_global.gfxExtensions = true;
FLA
Scaleform Player
Scaleform
SDK
Scaleform Player
Scaleform
API
Scaleform
API

```



```

        Scaleform
RIGHT, MIDDLE      (      16  )
        .
        Scaleform
        .
Flash Mouse Class
        .

```

## 2.1

```

Wii
3.2      4      . Gfx::Movie::SetMouseCursorCount(unsigned n)
        .
        Gfx::MouseEvent      Gfx::Movie::HandleEvent
Scaleform      . Gfx::MouseEvent
Gfx::MouseEvent  0

```

## 2.2

```

onRollOver, onRollOut, onDragOver, onDragOut, onPress, onRelease, onReleaseOutside
Scaleform
        0
        1
        1
        .
mc.onPress = function(mouseIdx:Number)
{
    if (mouseIndex == 0)
    else if (mouseIndex == 1)
    . . .
}

```



```

onPress    onRelease
.          1,    /    0
.
.

mc.onPress = function(mouseIdx:Number, keyboardOrMouse:Number)
{
    if (keyboardOrMouse == 0)
        . . .
    else
        . . .
    . . .
}

    onRollOver/Out    onDragOver/Out
rollover/dragover
.
    onRollOver/onRollOut    onDragOver/onDragOut
. onRollOver/onDragOver
roll
over    onRollOut/onDragOut
.
,    onRollOver/onRollOut
onDragOver/onDragOut    (
0    .    onRollOver/onDragOver
0
onRollOver/onRollOut    1
    onRollOut/onDragOut    1
onRollOut/onDragOut    0
,    onRollOver/onRollOut    onDragOver/onDragOut
.

mc.onRollOver = function(mouseIdx:Number, nestingIdx:Number)
{
    // do roll over animation only if nestingIdx == 0
    if (nestingIdx == 0)
        doOverAnimation();
    . . .
}
mc.onRollOut = function(mouseIdx:Number, nestingIdx:Number)
{

```

```

        // do roll out animation only if nestingIdx == 0
        if (nestingIdx == 0)
            doOutAnimation();
        . . .
    }

```

On(rollOver), on(rollOut)

onPress, onRelease

- onPressAux
- onReleaseAux
- onReleaseOutsideAux
- onDragOver
- onDragOut

, index != 0

(index == 0).

```

mc.onPressAux = function(mouseIdx:Number, keyboardOrMouse:Number, buttonIdx:Number)
{
    if (buttonIdx == 1) //
        . . .
    . . .
}

```

## 2.3 Mouse

- `onMouseMove = function() { }`
- `onMouseDown = function() { }`
- `onMouseUp = function() { }`
- `onMouseWheel = function(delta : Number, targetPath : String) { }`

```
(
    _global.gfxExtensions    true
    Scaleform
```

Scaleform RIGHT, MIDDLE

LEFT

6

```

    Mouse.addListener(mouseListener);

    /
    (
    .
    Mouse["LEFT"], Mouse["RIGHT"]
    Mouse["MIDDLE"] (Mouse["LEFT"]
    2.0 Mouse.LEFT ).
    .
    'undefined'
    'undefined'
    . 'undefined' LEFT
    .
    .

    ▪ onMouseMove = function([mouseIdx : Number],
        [x : Number], [y : Number]) { }
    ▪ onMouseDown = function(button : Number, [targetPath : String],
        [mouseIdx : Number], [x : Number], [y : Number],
        [dblClick : Boolean]) { }
    ▪ onMouseUp = function(button : Number, [targetPath : String],
        [mouseIdx : Number], [x : Number], [y : Number]) { }
    ▪ onMouseWheel = function(delta : Number, targetPath : String,
        [mouseIdx : Number], [x : Number], [y : Number]) { }

    mouseIdx 0
    x y .(_root ).
    onMouseDown dblClick . true

```

## 2.4

- *Arrow* -
- *Hand* - useHandCursor true
- *I-Beam* -

, Scaleform

C++ API

. C++                    Gfx::StateBag::SetUserEventHandler

FxPlayer.cpp

SDK                    *Mouse fla/swf*

'Cursor\_M'

"hand", "arrow"    'ibeam"

\_x    \_y

Scaleform            Mouse.setCursorType

cursorType    mouseX

cursorType            (Mouse["ARROW"], Mouse["HAND"],

Mouse["IBEAM"])

```
_global.gfxExtensions = 1;
// Hide system cursor.
Mouse.hide();

var mouseListener:Object = new Object;
mouseListener.onMouseMove = function(mouseIdx, x,y)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    if (x == undefined)
    {
        x = _xmouse;
        y = _ymouse;
    }
}
```

```

    }
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));

    cmc._x = x;
    cmc._y = y;
}
Mouse.addListener(mouseListener);

Mouse["setCursorType"] = function(cursorType, mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    switch(cursorType)
    {
        case Mouse["HAND"]:
            cmc.Cursor.gotoAndPlay("hand");
            break;
        case Mouse["ARROW"]:
            cmc.Cursor.gotoAndPlay("arrow");
            break;
        case Mouse["IBEAM"]:
            cmc.Cursor.gotoAndPlay("ibeam");
            break;
        default: return;
    }
}

// Override Mouse.show and Mouse.hide functions so
// that they affect our cursor.
ASSetPropFlags(Mouse, "show,hide", 0, 7);

Mouse.show = function(mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    cmc._visible = true;
}
Mouse.hide = function(mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    cmc._visible = false;
}

```

```

      .
      2.0
      .
      2.0
: Mouse["setCursorType"](...).

```

```
public function getButtonsState(mouseIndex : Number) : Number
```

Scaleform version: 2.2

```
mouseIndex : Number - 0
```

```
public function getTopMostEntity([mouseIndex : Number]) : Object
public function getTopMostEntity(testAll : Boolean,
                                  [mouseIndex : Number]) : Object
public function getTopMostEntity(x : Number, y : Number) : Object
public function getTopMostEntity(x : Number, y : Number, testAll: Boolean) : Object
```

Scaleform version: 1.2.34, 2.2

(onPress, onMouseDown, onRollOver )

```
MovieClip.hitTest      .      hitTest    x,y
      , getTopMostEntity      x,y
.      Mouse.getTopMostEntity(x, y).hitTest(x, y, true) == true.
```

mouseIndex : Number - 0

```
testAll : Boolean - (false) (true) .
getTopMostEntity 'true' .
```

```
x : Number, y : Number -
```

```
var listenerObj = new Object;
listenerObj.onMouseMove = function()
{
    var target = Mouse["getTopMostEntity"]();
    trace("Mouse moved, target = "+target);
}
Mouse.addListener(listenerObj);
var target = Mouse["getTopMostEntity"](480, 10);
trace("The character at the (480, 10) is " + target);
```

```
public function getPosition(mouseIndex : Number) : flash.geom.Point
```

Scaleform version: 2.2

```
_root .
flash.geom.Point .
```

```
mouseIndex : Number - 0
```

```
public function setCursorType(cursorType : Number, [mouseIndex : Number]) : void
```

Scaleform version: 1.2.32

```
cursorType . "
```



`cursorType` : Number – Mouse.ARROW, Mouse.HAND, Mouse.IBEAM

`mouseIndex` : Number – 0

hitTestDisable:Boolean [read-write]

Scaleform version: 2.1.51

```
true      MovieClip.hitTest      . ,
      .
      false .
      :
      TextField.hitTestDisable
      MovieClip.hitTestDisable
```

topmostLevel:Boolean [read-write]

Scaleform version: 2.1.50

```
true      .
      false .
      "topmostLevel "      :
Scaleform 3.0.71      "topmostLevel "
      true      . (      "topmostLevel"
      . )
Scaleform 3.0.72
      :      ObjectA      ObjectB      "topmostLevel"
```



```
public hitTest(x:Number, y:Number, [shapeFlag:Boolean],
               [ignoreInvisibleChildren:Boolean]):Boolean
```

Scaleform version: 2.1.51

```
3          hitTest
          . hitTest      x, y
( , _visible    false    _alpha    0          invisible
    )          "true"    .

ignoreInvisibleChildren:Boolean -          true
.
```

```
hitTestDisable:Boolean [read-write]
```

Scaleform version: 1.2.32

```
true          MovieClip.hitTest
          .
          .
false .
:
TextField.hitTestDisable
Button.hitTestDisable
```

```
noAdvance : Boolean
```

Scaleform version: 2.1.52

```
true                                     Advance    off      .  
                                     .         advance    (      ,  
                                     ).         "true"      Scaleform  
                                     .  
                                     :  
                                     _global.noInvisibleAdvance
```

topmostLevel:Boolean [read-write]

Scaleform version: 2.1.50

```
    true                                     .  
                                     .  
    false .  
                                     "topmostLevel " , :  
Scaleform 3.0.71                       "topmostLevel "  
    true                                     . ( "topmostLevel"  
                                     . )  
Scaleform 3.0.72  
    . : ObjectA ObjectB "topmostLevel"  
true ObjectA ObjectB  
    .  
    : "topmostLevel" swapDepth  
    .  
    :
```

```
TextField.topmostLevel  
Button.topmostLevel
```

```
rendererString:String [read-write]
```

Scaleform version: 2.2.55

```
MovieClip
```

```
myMovieInstance.rendererString = "SHADER_Blur"
```

```
rendererFloat:Number [read-write]
```

Scaleform version: 2.2.55

```
MovieClip
```

```
Float
```

```
myMovieInstance.rendererFloat = 4; // eg: C++ enum value
```

```
disableBatching:Boolean
```

Scaleform version: 4.0.17

focusGroupMask : Number

Scaleform version: 3.3.84

```
        focus group          ,          controller      focus
        . Focus group      setController      FocusGroup
        controller
        , "button1"  controller 0    1          focus
        focus group  controller
```

```
Selection.setControllerFocusGroup(0, 0);
Selection.setControllerFocusGroup(1, 1);
```

```
button1.focusGroupMask = 0x1; // bit 0 - focus group 0
movieclip2.focusGroupMask = 0x1 | 0x2 // bits 0 and 1 - focus groups 0 and 1
```

```
"focusGroupMask"          movieclip
```

Scaleform NetStream

```
onMetaData = function(info:Object) { }
```

Scaleform version: 3.0.63

, 2

onMetaData 2

1. audioTracks –

audioTracks

channelsNumber – (1-mono, 2- stereo, 6 – 5.1 surround sound)

totalSamples –

trackIndex – NetStream.audioTrack  
.( )

sampleRate –

2. subtitleTracksNumber –

audioTrack:Number [read-write]

Scaleform version: 3.0.63

set/get



```

var ns:NetStream = new NetStream(nc);
var audioTracks;

ns.onMetaData = function(info:Object)
{
    audioTracks = info.audioTracks;
}

if (audioTracks != undefined && audioTracks.length > 0)
    ns.audioTrack = audioTracks[0].trackIndex;

```

subtitleTrack:Number [read-write]

Scaleform version: 3.0.63

set . 0 .

```
onSubtitle = function(msg:String) {}
```

Scaleform version: 3.0.63

```

var ns:NetStream = new NetStream(nc);
var subtitlesNumber = 0;

ns.onMetaData = function(info:Object)
{
    subtitlesNumber = info.subtitleTracksNumber;
}

ns.onSubtitle = function(msg:String)
{
    sbTextField.text = msg;
}

if (subtitlesNumber > 0)
    ns.subtitleTrack = 1;

```

```
public function setNumberOfFramePools(pools : Number) : Void
```

Scaleform version: 3.0.68

```

        .
        , "frame pool" . CPU . , frame
pool . 1 .
        .( , NetStream.play() ).

```

```
public function setReloadThresholdTime(reloadTime : Number) : Void
```

Scaleform version: 3.0.68

```

        . Scaleform Video
        threshold( ) .
threshold( ) bitrate , reload
        . reload 0.8( ) .
        . , Netstream 2
        (NetStream.setBufferTime( ) threshold( ) 1 ,
1
        .
        . ,
        .
        .( , NetStream.play() )

```



Selection

Scaleform Selection 1.0

Selection.captureFocus() Selection

2.0

. ActionScript 2.0

Selection["captureFocus"]().

alwaysEnableArrowKeys : Boolean

Scaleform version: 1.2.34

"\_focusrect" false

( ). "\_focusrect = false;"

alwaysEnableArrowKeys "true"

Selection["alwaysEnableArrowKeys"] = true;

alwaysEnableKeyPress : Boolean

Scaleform version: 3.0.63

"\_focusrect" false( , )

onPress/ onRelease . "\_focusrect = false;"

alwaysEnableKeyPress true

```
Selection["alwaysEnableKeyPress"] = true;
```

```
public function captureFocus([doCapture:Boolean, controllerIdx:Number]) : void
```

Scaleform version: 1.2.34

```
doCapture true ( ) captureFocus
```

```
doCapture false captureFocus  
on
```

```
doCapture:Boolean - , /  
captureFocus true  
controllerIdx:Number - , capture /  
controller 0
```

```
Selection["captureFocus"](); // same as passing "true"  
Selection["captureFocus"](false);
```

disableFocusAutoRelease : Boolean

Scaleform version: 1.2.34

```
(
    ).
)
true
.
```

```
:
    Selection["disableFocusAutoRelease"] = true;
```

disableFocusKeys : Boolean

Scaleform version: 2.2.58

```
(TAB, Shift-TAB
).
```

```
Selection["disableFocusKeys"] = true;
```

```
moveFocus()
```

disableFocusRolloverEvent : Boolean

Scaleform version: 2.0.37

```

/
/
.
true
:

```

```
Selection["disableFocusRolloverEvent "] = true;
```

```
modalClip : MovieClip
```

```
Scaleform version: 2.2.58
```

```
                                "modal"
                                .
(TAB, Shift-TAB                )
                                ,
                                "
                                "
                                .
```

```
Selection["modalClip"] = _root.mc;
...
Selection["modalClip"] = undefined;
```

```
disableFocusKeys
moveFocus()
```

```
public function moveFocus(keyToSimulate : String [, startFromMovie:Object,
    includeFocusEnabledChars : Boolean = false,
    controllerIdx :Number]) : Object
```

```
Scaleform version: 2.2.58
```

```
(TAB, Shift-TAB                )
                                .disableFocusKeys modalClip
                                .
```

keyToSimulate:String – "up", "down", "left", "right", "tab", "shifftab"

startFromMovie:Object –

moveFocus . null

undefined .

includeFocusEnabledChars:Boolean –focusEnabled ,

tabEnabled/tabIndex moveFocus

0

tabEnabled/tabIndex

controllerIdx:Number – , capture /  
controller 0 .

```
Selection["moveFocus"]("up");  
Selection["moveFocus"]("tab", _root.mc);  
Selection["moveFocus"]("tab", _root.mc, true);  
Selection["moveFocus"]("tab", null, true);
```

undefined .

disableFocusKeys  
modalClip



```

public function findFocus(keyToSimulate : String [, parentMovie:Object, loop :
    Boolean, startFromMovie:Object, includeFocusEnabledChars : Boolean,
    controllerIndex : Number]) : Object

```

Scaleform version: 3.3.84

```

    TAB, Shift + TAB                                focus
    . disableFocusKeys    setModalClip/getModalClip
    focus
    .

keyToSimulate:String - ( "up", "down", "left", "right", "tab",
    "shift + tab")
parentMovie - modal clip    movie clip    . focus
    clip    . null
loop - focus    Boolean
    . , focus
    "down" , findFocus "null"( "false"
    ) focus (
    "true" )
startFromMovie:Object - focus    findFocus
    null    focus

includeFocusEnabledChars:Boolean - tabEnabled/tabIndex
    focusEnabled
    moveFocus    false
    tabEnabled/tabIndex
    focus
controllerIndex - focus    controller zero base index
    focus group    controller focus
    .

```

```
var a = Selection["findFocus"]("up");
```

```
, focus      null
```

```
:
  disableFocusKeys
  setModalClip
  getModalClip
```

```
public function setModalClip(modalClip : Object, controllerIndex : Number)
```

Scaleform version: 3.3.84

```

      movie clip    focus      "modal" clip
TAB, Shift + TAB    "TAB      "      movie clip
focus              .

```

```

modalClip      - A modal clip.
controllerIndex - controller  zero base index

```

```
public function getModalClip(controllerIndex : Number) : Object
```

Scaleform version: 3.3.84

```

controller      modal clip

```

```

controllerIndex - controller  Zero base index

```

modal clip      undefined      .

```
public function setControllerFocusGroup (controllerIndex : Number, focusGroupIdx :
                                         Number) : Boolean
```

Scaleform version: 3.3.84

```

        controllerIndex      controller  focus group      .
        controller  focus group 0      ,      focus
        .      ,      controller      focus      .
        ,      controller      focus      (      )
setControllerFocusGroup (1,1)  controller 1      focus group      .
setControllerFocusGroup (1,0)      controller 0  1      focus
.
```

controllerIndex      – controller      zero base index      .  
focusGroupIdx      – focus group      zero base index      .

```
Selection["setControllerFocusGroup"](0,0);
Selection["setControllerFocusGroup"](1,1);
Selection["setControllerFocusGroup"](2,1);
```

true      .

```
public function getControllerFocusGroup (controllerIndex : Number) : Number
```

Scaleform version: 3.3.84

controller focus group index .

controllerIndex – controller zero base index .

focus group zero base index .

```
public function getFocusArray(mc : Object) : Array
```

Scaleform version: 3.3.84

movie clip/button/text field focus controller  
index .

mc - movie clip, button text field.

zero base index ( ).

```
public function getFocusBitmask(mc : Object) : Number
```

Scaleform version: 3.3.84

movieclip/button/textfield , bit  
focus controller .

mc - movie clip, button text field.

controller .

public function getControllerMaskByFocusGroup (focusGroupIdx : Number) : Number

Scaleform version: 3.3.84

controller , focus group  
. setControllerFocusGroup  
.

focusGroupIdx - focus group index .

controller .

numFocusGroups : Number

Scaleform version: 3.3.84

setControllerFocusGroup focus group . focus  
group 0 3 , numFocusGroups 2 .

## 7 TextField

Scaleform    TextField

```
8                    TextField                    .
9                    .                    IME,                    ,
                  (                    ,                    ),                    ,
                  .
```

### 7.1

                  .

autoFit:Boolean    [read-write]

Scaleform version: 2.0.39

                  .

                  false    .

public function appendText(newText:String):void

Scaleform version: 2.0.37

newText                    .

                  (+=)                    (    , my\_txt.text += appendingText)

                  .

                  :                    style sheet    text field                    .

```
newText:String -
```

```
:
```

```
appendHtml()
```

```
public function appendHtml(newHtml:String):void
```

Scaleform version: 2.0.37

```
newHtml          HTML
```

```
.
```

```
htmlText          (+=)          ( , txt.htmlText += moreHtml).
```

```
htmlText          += HTML
```

```
HTML
```

```
HTML
```

```
.
```

```
HTML
```

```
,
```

```
newHtml
```

```
HTML
```

```
(
```

```
newHtml
```

```
HTML
```

```
+=
```

```
).
```

```
.
```

```
:
```

```
style sheet text field
```

```
.
```

```
newHtml:String -
```

```
HTML
```

```
:
```

```
appendText()
```

```
caretIndex:Number [read-only]
```

Scaleform version: 2.0.37

```
(  
(  
caret  
caret  
Selection.getCaretIndex()  
Selection.getCaretIndex() -1  
Caret 0 ( 0 ).
```

```
Selection.getCaretIndex()
```

```
public function getCharBoundaries(charIndex:Number): flash.geom.Rectangle
```

Scaleform version: 2.0.37

```
.  
,  
(  
, 'a', 'w' 'k'  
).
```

*Baywio*ks

```
charIndex:Number - 0 (  
0, 1)
```



```
flash.geom.Rectangle - x y ,
. , (0,0) .
```

```
getExactCharBoundaries()
```

```
public function getExactCharBoundaries(charIndex:Number): flash.geom.Rectangle
```

Scaleform version: 2.0.37

```
.
(getCharBoundaries() ).
'a', 'w' 'k'
.
```

*Baywio*ks

```
charIndex:Number - 0 (
0, 1)
```

```
flash.geom.Rectangle - x y ,
. , (0,0) .
```

```
getCharBoundaries()
```

```
public function getCharIndexAtPoint(x:Number, y:Number): Number
```

Scaleform version: 2.0.37

x y (0 ).

x:Number – x

y:Number – y

Number - 0 ( 0,  
1). -1 .

```
public function getFirstCharInParagraph(charIndex:Number): Number
```

Scaleform version: 2.0.37

charIndex .

charIndex:Number - 0 ( 0,  
1)

Number – 0

```
public function getLineIndexAtPoint(x:Number, y:Number): Number
```

Scaleform version: 2.0.37

```

    x    y                                0
    .

    x:Number -      x
    y:Number -      y

    Number - 0                                (      0,
    1).                                -1      .

```

```
public function getLineLength(lineIndex:Number): Number
```

Scaleform version: 2.0.37

```

    .

    lineIndex:Number - 0                                (      0,
    1)

    Number -

```

```
public function getLineMetrics(lineIndex:Number): Object
```

Scaleform version: 2.0.43

ascent : Number

ascent

descent : Number

descent

height : Number

( )

leading : Number

width : Number

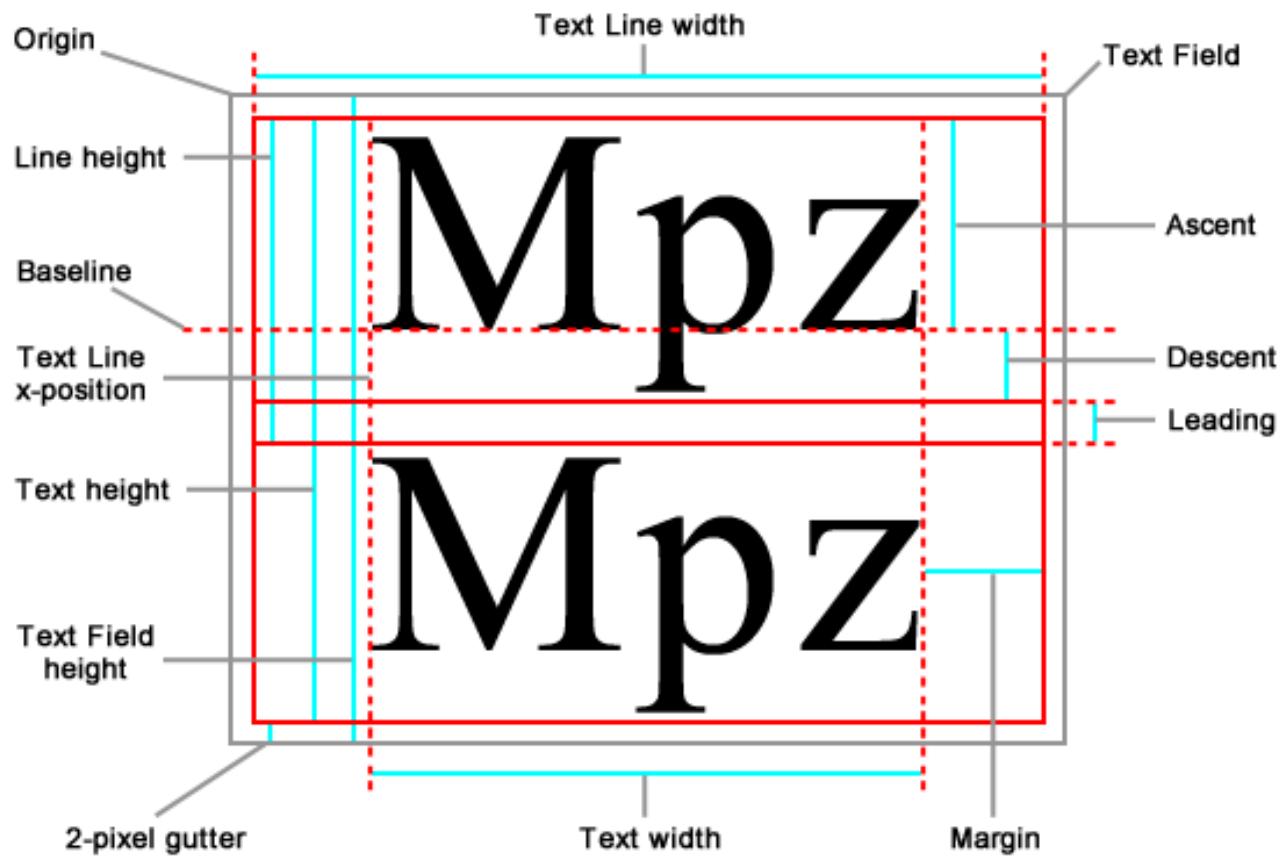
width

(

).

x : Number

x



lineIndex: Number - 0 ( 0,  
1)

Number –

```
public function getLineOffset(lineIndex: Number): Number
```

Scaleform version: 2.0.37

lineIndex .

```
lineIndex:Number - 0 ( 0,
1)
```

Number - 0

```
public function getLineText(lineIndex:Number): String
```

Scaleform version: 2.0.43

```
lineIndex .
```

```
lineIndex:Number - 0 ( 0,
1)
```

String –

```
hitTestDisable:Boolean [read-write]
```

Scaleform version: 1.2.32

```
true MovieClip.hitTest . ,
```

```
false .
```

```
noTranslate:Boolean [read-write]
```

Scaleform version: 1.2.34

true GfxTranslator::Translate .

false .

numLines:Number [read-only]

Scaleform version: 2.0.43

. wordWrap true

wrap .

topmostLevel:Boolean [read-write]

Scaleform version: 2.1.50

true .

.

false .

"topmostLevel " , ::

• Scaleform 3.0.71 "topmostLevel " true . ( "topmostLevel" . )

Scaleform 3.0.72

ObjectA ObjectB "topmostLevel"  
true ObjectA ObjectB

: "topmostLevel" swapDepth

:  
MovieClip.topmostLevel  
Button.topmostLevel

focusGroupMask : Number

Scaleform version: 3.3.84

focus group , controller focus  
. Focus group setController FocusGroup  
controller  
 , "button1" controller 0 1 focus  
focus group controller

```
Selection.setControllerFocusGroup(0, 0);  
Selection.setControllerFocusGroup(1, 1);
```

```
button1.focusGroupMask = 0x1; // bit 0 - focus group 0  
movieclip2.focusGroupMask = 0x1 | 0x2 // bits 0 and 1 - focus groups 0 and 1
```

"focusGroupMask" movieclip

## 7.2



alwaysShowSelection:Boolean [read-write]

Scaleform version: 2.1.45

```

        Scaleform
        true
        Scaleform
        (" "
        ).
        false
        :
        selectionBkgColor
        selectionTextColor
        inactiveSelectionBkgColor
        inactiveSelectionTextColor

```

```

public function copyToClipboard([richTextClipboard:Boolean], [startIndex:Number],
[endTime:Number]):void

```

Scaleform version: 2.1.45

```

        richClipboard:Boolean – true
        useRichTextClipboard
        startIndex:Number –
        selectionBeginIndex
        endIndex:Number –
        selectionEndIndex
        :

```

```

useRichTextClipboard
selectionBeginIndex
selectionEndIndex
cutToClipboard()
pasteFromClipboard()

```

```

public function cutToClipboard([richClipboard:Boolean], [startIndex:Number],
[endTime:Number]):void

```

Scaleform version: 2.1.45

```

richClipboard:Boolean - true
                                useRichTextClipboard
startIndex:Number -
                                selectionBeginIndex
endTime:Number -
                                selectionEndIndex
:
useRichTextClipboard
selectionBeginIndex
selectionEndIndex
copyToClipboard()
pasteFromClipboard()

```

```

inactiveSelectionBkgColor:Number [read-write]

```

Scaleform version: 2.1.45

```
        . alwaysShowSelection      true
        . 0xAARRGGBB               , AA      [0...255]    16      , RR
        [0...255]    16      , GG      [0...255]    16      , BB
[0...255]    16      .
        0
        .
        . 0xFFFF0000(
0xFF(255))      backgroundColor      0xFF0000
:
alwaysShowSelection
inactiveSelectionTextColor
selectionBkgColor
selectionTextColor
```

inactiveSelectionTextColor: Number [read-write]

Scaleform version: 2.1.45

```
        . alwaysShowSelection      true
        . 0xAARRGGBB               , AA      [0...255]    16      , RR
        [0...255]    16      , GG      [0...255]    16      , BB
[0...255]    16      .
        0
        .
        . 0xFFFF0000(
0xFF(255))      textColor      0xFF0000
:
alwaysShowSelection
```

inactiveSelectionBkgColor  
selectionBkgColor  
selectionTextColor

noAutoSelection:Boolean [read-write]

Scaleform

selectionBeginIndex: Number [read-only]

Scaleform version: 2.1.45

```
0
caretIndex
Selection.getBeginIndex() selectionBeginIndex
Selection.getBeginIndex() -1
:
caretIndex
selectionEndIndex
Selection.getBeginIndex()
```

selectionEndIndex: Number [read-only]

Scaleform version: 2.1.45

```
0
caretIndex
Selection.getEndIndex() selectionEndIndex
Selection.getEndIndex() -1
:
caretIndex
selectionBeginIndex
Selection.getEndIndex()
```

selectionBkgColor: Number [read-write]

Scaleform version: 2.1.45

```
        . alwaysShowSelection      true
        . 0xAARRGGBB                , AA      [0...255]    16      , RR
        [0...255]    16      , GG      [0...255]    16      , BB
[0...255]    16      .
        0
        .
        0xFFFF0000(
0xFF(255))      backgroundColor      0xFF0000
:
inactiveSelectionTextColor
inactiveSelectionBkgColor
selectionTextColor
```

selectionTextColor: Number [read-write]

Scaleform version: 2.1.45

```
        . alwaysShowSelection      true
        . 0xAARRGGBB                , AA      [0...255]    16      , RR
        [0...255]    16      , GG      [0...255]    16      , BB
[0...255]    16      .
        0
        .
        0xFFFF0000(
0xFF(255))      textColor      0xFF0000
```

```

:
inactiveSelectionBkgColor
inactiveSelectionBkgColor
selectionBkgColor

```

```
useRichTextClipboard:Boolean [read-write]
```

Scaleform version: 2.1.45

```

        . True      Scaleform
        (  ,      ).
        useRichTextClipboard  true
        false .

```

## 7.3

```

        .      (  ,
        ) Scaleform
        .

```

```
fontScaleFactor:Number [read-write]
```

Scaleform version: 2.0.43

```

fontScaleFactor
1.0 .

```

textAutoSize:String [read-write]

Scaleform version: 2.0.43

textAutoSize

"none", "shrink" "fit" ("shrink" "fit")

( 5pt)

textAutoSize "fit"

"shrink"



"none"

verticalAlign:String [read-write]

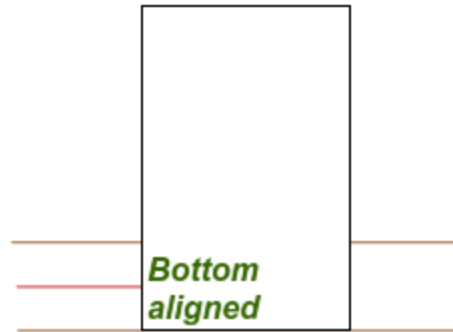
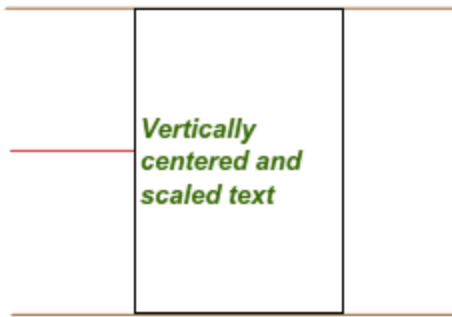
Scaleform version: 2.0.43

"none" ( "none"

"top"), "bottom", "center" "center"

, "bottom" ( ).





"none" .

:

verticalAutoSize  
TextField.align

verticalAutoSize:String [read-write]

Scaleform version: 2.0.43

verticalAutoSize "none" ( ), "top",  
"bottom", "center" . verticalAutoSize "none" ( )

verticalAutoSize "top" wordWrap true  
TextFieldAutoSize ( ).

verticalAutoSize "center"  
( ).

verticalAutoSize "bottom"  
( ).



"none" .

:  
verticalAlign  
TextField.autoSize

## 7.4 HTML

Scaleform HTML

ALPHA="#xx"

00 FF (0 – 255). 0 , 255  
. COLOR ALPHA HTML

.

Scaleform 2.0/2.1

<IMG>  
. Scaleform ("http://", "file://"  
) . 5.6

.

.

.

## 7.5 Shadow

( , ) .

```
blurX:Number [read-write]
blurY:Number [read-write]
```

Scaleform version: 2.0.41

. 0 15.9 ( ).

0 .

:

blurStrength

```
blurStrength:Number [read-write]
```

Scaleform version: 2.0.41

. 0 15.9 ( ).

0 .

:

blurX  
blurY

```
shadowAlpha:Number [read-write]
```

Scaleform version: 2.0.41

. 0.00 1.00 . .25  
 25%  
 :  
 shadowColor

shadowAngle:Number [read-write]

Scaleform version: 2.0.41

DropShadowFilter . 0  
 ) . 45 .  
 :  
 DropShadowFilter

shadowBlurX:Number [read-write]  
 shadowBlurY:Number [read-write]

Scaleform version: 2.0.41

. 0 15.9 ( ).  
 0 .  
 :  
 shadowStrength

shadowColor: Number [read-write]

Scaleform version: 1.1.31

( shadowStyle ). 0xRRGGBB .  
RR 16 [0..255], BB 16 [0..255] GG  
16 [0..255] .

shadowStyle

shadowDistance: Number [read-write]

Scaleform version: 2.0.41

DropShadowFilter .

:  
DropShadowFilter

shadowHideObject: Boolean [read-write]

Scaleform version: 2.0.41

. True .  
. false ( ) .

shadowKnockOut:Boolean [read-write]

Scaleform version: 2.0.41

(true) .  
false ( ) .

shadowQuality:Number [read-write]

Scaleform version: 2.0.41

. 1 2 . 1  
, 2 .

shadowStrength:Number [read-write]

Scaleform version: 2.0.41

. 0 15.9 ( ).  
0 .  
:  
shadowBlurX  
shadowBlurY

shadowStyle:String [read-write]

Scaleform version: 1.1.31

shadowColor . shadowStyle

field.shadowStyle = "s{0,-1.5}{-1.4,1.2}{1.4,1.2}t{0,0}";  
field.shadowColor = 0xff0000;

"s" shadowColor  
"t" TextField.textColor  
(delimiter)  
"t"  
"t{0,0}"

draw primitive

:  
shadowColor  
TextField.textColor

## 7.6



```

public setImageSubstitutions(substInfoArr:Array) : void
public setImageSubstitutions(substInfo:Object) : void
public setImageSubstitutions(null) : void

```

Scaleform version: 2.0.38

SWF

SWF

- 1.
2. ( ) ( )CTRL
- Linkage
3. Export for ActionScript ( , myImage)
4. OK

BitmapData

```

import flash.display.BitmapData;
var imageBmp:BitmapData = BitmapData.loadBitmap("myImage");

```

```

!(import flash.display.BitmapData;
flash.display.BitmapData ).
"undefined"

```

ID

subString:String

image : BitmapData

width : Number

height : Number

baseLineY : Number

Y

(

)

id : String

"updateImageSubstitution"

id

"substInfoArr"

"substInfo"

setImageSubstitutions(substInfo:Object)

setImageSubstitutions(substInfoArr:Array)

setImageSubstitutions

setImageSubstitutions(null)

setImageSubstitutions

substInfoArr:Array -

( )

substInfo:Object -

( ).

null -

:

updateImageSubstitution()

:

```
var b1 = BitmapData.loadBitmap("smile1");
var b2 = BitmapData.loadBitmap("smile2");
var b3 = BitmapData.loadBitmap("smile3");
var a = new Array;
a[0] = { subString:"=", image:b1, baseLineY:35, width:20, height:20, id:"sm=" } };
a[1] = { subString":-)", image:b2, baseLineY:20, id:"sm:-)" } };
a[2] = { subString":\\", image:b3, baseLineY:35, height:100 } };
a[3] = { subString":-\\", image:b1 } };
t.setImageSubstitutions(a);
```

"=")

"smile1"

.

public updateImageSubstitution(id:String, image:BitmapData) : void

Scaleform version: 2.0.38

setImageSubstitutions

.

id:String - ID, setImageSubstitutions

id

image:BitmapData - . Null

:

setImageSubstitutions()

:

t.updateImageSubstitution("sm=", b3);

. onEnterFrame

setInterval

. updateImageSubstitution

. ,

.

```

var phase = 0;
var b1a = BitmapData.loadBitmap("smile1a");
var b2a = BitmapData.loadBitmap("smile2a");

onEnterFrame = function()
{
    if (phase % 10 == 0)
    {
        if (phase % 20 == 0)
        {
            _root.t.updateImageSubstitution("sm=", b1);
            _root.t.updateImageSubstitution("sm:-)", b2);
        }
        else
        {
            _root.t.updateImageSubstitution("sm=", b1a);
            _root.t.updateImageSubstitution("sm:-)", b2a);
        }
    }
    ++phase;
}

```

## 7.7 IME

IME

.

disableIME:Boolean [read-write]

Scaleform version: 2.1.50

True

IME

.

false .

public function getIMECompositionStringStyle(categoryName:String): Object

Scaleform version: 2.1.50

```

IME
                                . categoryName
                                setIMECompositionStringStyle
                                .

categoryName:String – IME
                                (setIMECompositionStringStyle
                                ).

Object –
                                (\
                                ).

:
setIMECompositionStringStyle
disableIME

public function setIMECompositionStringStyle(categoryName:String,
                                styleDescriptor:Object): Number

```

Scaleform version: 2.1.50

```

IME
                                . categoryName
                                .

"compositionSegment" –
"clauseSegment"–
"convertedSegment" –
"phraseLengthAdj" –
"lowConfSegment" –

styleDescriptor
                                .

textColor : Number

backgroundColor : Number

```

underlineColor : Number

underlineStyle : String

"single"

"thick"

"dotted"

"ditheredSingle"

"ditheredThick"

0xRRGGBB

RR

16

[0..255],

BB

16

[0..255]

GG

` 16

[0..255]

categoryName:String – IME

( )

styleDescriptor:Object –

( )

:

getIMECompositionStringStyle

disableIME

TextFormat , , .

alpha:Number [read-write]

Scaleform version: 2.0.41

. 0 – 100 (%) .

TextFormat.color

.

:

TextFormat

HTML <FONT ALPHA='xx'>

Stage

Scaleform

```
. "visibleRect", "safeRect" "originalRect"  
 , "onResize"
```

```
1.0 Stage.visibleRect
```

```
2.0
```

ActonScript 2.0

```
Stage["visibleRect"]..
```

```
visibleRect:flash.geom.Rectangle [read-only]
```

Scaleform version: 2.2.56

```
:  
safeRect  
originalRect
```

```
safeRect:flash.geom.Rectangle [read-only]
```

Scaleform version: 2.2.56

```
. GFX::Movie::SetSafeRect  
 "visibleRect"  
:  
visibleRect
```



originalRect

originalRect:flash.geom.Rectangle [read-only]

Scaleform version: 2.2.56

```

        SWF                                SWF                                .
        "Document properties"              (600px by 400px)                  {0, 0,
{600, 400} }                               .
:
visibleRect
safeRect
```

onResize = function([visibleRect:flash.geom.Rectangle]) {}

Scaleform version: 2.2.56

```

onResize                                .
    Stage.visibleRect                    .
:
visibleRect
```

public function translateToScreen(pt:Object): Point

Scaleform version: 3.3.84

```

Stage                                point                                .
```

pt:Object – point x, y Object .  
flash.geom.Point Object .

Point – point .

```

        Array.sort      Array.sortOn
        ,
        ,
        ,
        (umlaut, accent, acute ) z
        ['á', 'z', 'a']
        ['a', 'z', 'á']
        ,
        ['a', 'á', 'z']
        .

```

Array.LOCALE: Scaleform Array.sort/sortOn

```

        .
        Array.LOCALE
        .

```

```

var arr = ['á', 'z', 'a'];
var newArr1 = arr.sort(); // returns ['a', 'z', 'á']
var newArrLoc = arr.sort(Array.LOCALE); // returns ['a', 'á', 'z']
var newArrRev = arr.sort(Array.LOCALE | Array.DESENDING); // ['z', 'á', 'a']

```

```

var arr = ['Á', 'z', 'a'];
var a = arr.sort(Array.LOCALE | Array.CASEINSENSITIVE); // returns ['a', 'Á', 'z']

```

```

        Array.LOCALE
        .

```

Scaleform version: 3.0.65

```

:
String.localeCompare

```

2

0,

"

"

Array.LOCALE

```
public localeCompare(other:String [, caseInsensitive:Boolean]) : Number
```

Scaleform version: 3.0.65

other:String -

caseInsensitive:Boolean -

:

Array.sort

```
NetStream . Scaleform  
NetStream ,
```

```
clipRect: flash.geom.Rectangle [read-write]
```

```
Scaleform version: 3.0.70
```

```
:  
video.clipRect = new flash.geom.Rectangle(10, 20, 100, 100);
```

## 13

noInvisibleAdvance : Boolean

Scaleform version: 2.1.52

True

SWF

( , ). "true"

Scaleform

:

MovieClip.noAdvance

imecommand(command:String, parameters:String) : Void

Scaleform version: 2.1.50

fscommand

Scaleform IME

. Scaleform

.

public function getIMECompositionStringStyle(categoryName:String): Object

Scaleform version: 2.1.50

IME

setIMECandidateListStyle

.

Object –

```
:  
setIMECandidateListStyle  
TextField.disableIME
```

```
public function setIMECandidateListStyle(styleDescriptor:Object)
```

Scaleform version: 2.1.50

styleDescriptor

```
styleObject.textColor = 0x5EADFF;  
styleObject.selectedTextColor = 0xFFFFFFFF;  
styleObject.fontSize = 20;  
styleObject.backgroundColor = 0x001430;  
styleObject.selectedTextBackgroundColor = 0x2C5A99;  
styleObject.indexBackgroundColor = 0x12325A;  
styleObject.selectedIndexBackgroundColor = 0x2C5A99;  
styleObject.readingWindowTextColor = 0xFFFFFFFF;  
styleObject.readingWindowBackgroundColor = 0x001430;  
styleObject.readingWindowFontSize = 20;
```

textColor : Number

selectedTextColor : Number

fontSize : Number

backgroundColor : Number

seletedTextBackgroundColor : Number

indexBackgroundColor : Number

selectedIndexBackgroundColor : Number

readingWindowTextColor : Number

readingWindowBackgroundColor : Number

readingWindowFontSize : Number

0xRRGGBB . RR 16 [0..255],  
BB 16 [0..255] GG 16 [0..255] .  
:  
getIMECandidateListStyle  
TextField.disableIME



Scaleform 3Di    Scaleform    AS2    .    3Di  
ActionScript    .

Scaleform version: 3.2.82

Z ( )    .    0    .

Z    .    0    .

X( )    .    0    .

Y( )    .    0    .

16    float    (4 x 4)    3D    .    NULL  
3D    2D    .

(Field Of View)    ,    0    180  
FOV ( 55)    .

gfxExtensions    ActionScript    true

```
Key.getCode(controllerIdx:Number)
Key.getAscii(controllerIdx:Number)
Key.isDown(controllerIdx:Number)
Key.isToggled(controllerIdx:Number)
```

```

/
controllerIdx      0      .      Key      .      ,
```

```
Key.onKeyDown(controllerIdx:Number)
```

```
Key      onKeyDown      /
.
```

```
Selection.setFocus(ch:Object, controllerIdx:Number)
Selection.getFocus(controllerIdx:Number)
Selection.getBeginIndex(controllerIdx:Number)
Selection.getEndIndex(controllerIdx:Number)
Selection.setSelection(start:Number, end:Number, controllerIdx:Number)
Selection.getCaretIndex(controllerIdx:Number)
```

```

/
controllerIdx      0      .      Selection      .      ,
```

```
Selection.onSetFocus(old:Object, new:Object, controllerIdx:Number)
```

Selection                      onSetFocus                      /

.

```
MovieClip.onSetFocus(old:Object, controllerIdx:Number)
MovieClip.onKillFocus(new:Object, controllerIdx:Number)
Button.onSetFocus(old:Object, controllerIdx:Number)
Button.onKillFocus(new:Object, controllerIdx:Number)
TextField.onSetFocus(old:Object, controllerIdx:Number)
TextField.onKillFocus(new:Object, controllerIdx:Number)
TextField.onChangeed(controllerIdx:Number)
```

MovieClip/Button/TextField                      /

.

System.capabilities.numControllers –                      controller                      .